

Tentamen Databases 26 maart 2013 (12:00-14:45)

Grading: starts at 1; question 1(a) gives 1.5 points, 2(a) and 2(b) give 1 point; the rest 0.5 points. Please answer all questions either in English (preferred) or in Dutch.

If you did the partial exam this year, then we will use your score there, if it is larger than the total you obtain on questions (1) and (2) in this exam (i.e. you might skip questions 1+2 now if your partial exam score is good).

1 Conceptual Design

Consider the following case: *We are modeling a chain of pharmacies. Patients have a first name, last name, gender and birthdate. We typically know the address of a person. An address consists of a street, zipcode, and city, and we also keep an email address and phone number. For doctors we keep the same data as for patients but that additionally have a specialization. Each pharmaceutical company is identified by name and website URL. For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharmaceutical company, and the trade name identifies a drug uniquely from among the products of that company. Each pharmacy has a name, VAT number, address, and a manager. Every patient has a primary physician (i.e. doctor). Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another. A prescription made by a doctor for a patient. Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it. Pharmaceutical companies have long-term contracts with pharmacies. A pharmaceutical company can contract with several pharmacies, and a pharmacy can contract with several pharmaceutical companies. For each contract, you have to store a start date, an end date, and the text of the contract.*

Question 1(a): Provide a conceptual database model in the form of a UML Class Diagram. Explain the most important design choices and document relevant assumptions.

Question 1(b): Give the associated relational schema, indicating relations $R(A_1, \dots)$ with their attributes A_i , primary keys underlined, and foreign key relationships ($\rightarrow$). Separately also comment on NULLable attributes (if any), and constraints that a relational database could check, including candidate keys.

2 Normalization

Given $R(A, B, C, D, E, F)$, and FDs: $f = \{ AE \rightarrow DEF, C \rightarrow DE, F \rightarrow CD \}$

Show your intermediate steps in all the answers.

Question 2(a): is f canonical (a minimal basis)? If not, make it so.

Question 2(b): what are the minimal (candidate) keys?

Question 2(c): is R in BCNF already? If not, make it so, and tell if any FD's get lost.

Question 2(d): is R in 3NF already? If not, make it so.

3 SQL

Consider the following schema:

Suppliers(sid,sname,saddress)

Parts(pid,pname,color)

Catalog(sid→Suppliers,pid→Parts,cost)

The primary keys are underlined and → Rel indicates a foreign key relationship with the primary key of Rel.

Formulate the following queries in SQL. You will only obtain maximal points for your answers if your answers avoid use of GROUP BY in favor of existential quantification.

Question 3(a): Find the names of the cheapest suppliers of part 42 (i.e. pid=42).

Question 3(b): Find (pid,sid) of those suppliers and products who do not offer that product at lowest price.

4 Transactions

Question 4(a): Describe the notions of a serial, serializable and conflict equivalent and conflict serializable schedules.

Question 4(b): Consider the following concurrent schedule. Is this schedule conflict serializable? If it is, give an equivalent serial schedule.

T ₀	T ₁	T ₂
		read(Y)
		read(Z)
read (X)		
writeX)		
		write(Y)
		write(Z)
	read(Z)	
read(Y)		
write(Y)		
	read(Y)	
	write(Y)	
	read(X)	
	write(X)	

5 Database APIs

Question 5(a): describe three features of JDBC that are often used to tune the performance of applications.

Question 5(b): explain what the term “impedance mismatch” refers to and discuss LinQ in this respect.

6 Data Warehousing and NoSQL

Question 6(a): Using the SQL 1999 GROUP BY CUBE extension, and assuming productName has 10 different values and shopCity has 5 different values (and all combinations occur); how many tuples will be returned by the query:

```
SELECT shopCity, productName, sum(price_paid)
FROM sales
GROUP BY CUBE(shopCity,productName)
```

Question 6(b): Discuss some advantages and disadvantages of HBase (a key-value noSQL store).